

Toward Family-Centered AI Literacy: Extending CollabPEAS for Intergenerational Agent Design

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AI agents are now part of everyday household technologies, yet most families lack shared vocabulary for discussing how these systems work or why they behave the way they do. This paper presents CollabPEAS, a collaborative virtual reality (VR) system where pairs of learners jointly design a virtual cleaning robot using the PEAS framework (Performance measure, Environment, Actuators, Sensors), developing the ability to reason about how AI agent behavior is shaped by human design choices. CollabPEAS was built for peer dyads, but its collaborative structure, including alternating lead-reviewer roles, an agree-before-placing mechanism, and structured reflection prompts, translates naturally to parent-child pairs where one person knows more than the other. The paper reflects on what changes when the dyad is a family, what new design questions that raises, and how the system's collaboration features might double as rehearsal for the kind of ongoing AI governance that families increasingly need to practice together.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing**; **Virtual reality**; • **Applied computing** → *Collaborative learning*.

Additional Key Words and Phrases: AI literacy; family-centered AI; intergenerational learning; PEAS framework; collaborative VR; agent design

1 Introduction

AI agents are already part of the technologies families use every day: voice assistants answer children's questions, recommendation engines shape what the family watches and reads, and autonomous systems make decisions that affect the whole household. But families rarely have the vocabulary to talk about how these systems work, why they behave a certain way, or who made the choices that shaped them. This is not just a gap in individual understanding. Parents and children bring different mental models of AI into the same home [1], and the family has become an important but underexplored site for building shared AI literacy [1, 8].

Most AI literacy education happens in classrooms, where learners are assumed to start at roughly the same level [4]. Families are different. A parent might understand the goal-setting and institutional dimensions of an AI system, while a child is more comfortable with its interactive and computational aspects. Rather than treating this asymmetry as a problem, research on intergenerational technology learning suggests it can be a resource: families build richer shared understanding when roles shift fluidly between teacher and learner depending on the task [12].

This paper presents CollabPEAS, a collaborative VR system where pairs of learners jointly design a virtual cleaning robot using the PEAS framework (Performance measure, Environment, Actuators, Sensors) [7], learning to reason about how AI agents sense, decide, and act. The system was built for peer dyads in educational settings. The paper asks what happens when that dyad is a parent and child, what the family context demands differently, and how CollabPEAS might grow to support AI literacy as something families build together.

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2 Related Work

2.1 Family as a Site for AI Literacy

Druga et al. [1] studied how children and parents learn about AI together at home and found that families do have informal AI literacy conversations: asking what a voice assistant knows, debating whether a recommendation seems fair, or deciding how much to trust an autonomous system. But these conversations tend to be fragmented, the vocabulary differs across generations, and parents often feel unprepared to explain AI concepts they themselves find confusing. The family home is a third space for AI literacy, sitting between formal classroom instruction and isolated individual use, where learning is shaped by relationships and shared experience [1].

Recent work on parent-child collaborative AI systems has looked at how to structure these interactions productively. Sun et al. [8] found that parents working with AI-supported storytelling took on different roles depending on what the child needed at any given moment: sometimes co-author, sometimes observer, sometimes facilitator. That kind of flexibility produced richer engagement than assigning fixed roles from the start. Yuan et al. [12] found something similar: intergenerational pairs do better when the technology gives them a shared artifact both can look at, change, and talk about, even if they bring different kinds of knowledge to it.

2.2 Collaborative AI Literacy Systems

Existing AI literacy systems for non-technical learners have focused primarily on conceptual understanding through media-rich curricula, hands-on machine learning activities, and block-based coding environments [4, 5, 11]. These approaches teach *about* AI agents but rarely give learners the chance to practice the structured specification reasoning that connects design choices to observable behavior. CollabPEAS addresses this by using the PEAS framework [7] as a learner-facing scaffold that makes agent specification choices explicit, consequential, and jointly negotiated. Instead of simply using or observing an AI system, learners author one, which supports the competencies Long and Magerko [4] identify as central to AI literacy: Recognizing AI (C1), Decision-Making (C8), Action and Reaction (C14), Sensors (C15), and Programmability (C17).

3 CollabPEAS: System Overview

CollabPEAS is a multiplayer VR learning environment built in Unity 6 for Meta Quest headsets, using Photon Fusion 2 with Shared Authority topology for real-time synchronization. Both participants occupy the same virtual space with access to all shared objects. The system has two main components.

PEAS-Guided Design Flow. The learning experience is a five-step progressive design flow in which dyads collaboratively specify a household cleaning robot. Participants define a performance measure (“percentage of floor area cleared in 10 minutes”), characterize the apartment environment and select sensors, choose actuators, program agent behavior through block-based coding (SENSE → CONDITION → ACTION → CONTROL chains), and test, debug, and revise the agent. Each step makes the corresponding PEAS component spatially grounded and jointly inspectable in VR, and the system enforces compatibility between blocks so that decision-rule construction is accessible without programming prerequisites [3]. The flow is fully implemented.

Collaboration Design Supports. To scaffold productive joint specification, the system includes five collaboration supports: (1) *Agree-before-placing*, in which block placements enter a pending state requiring the partner’s approval before taking effect; (2) *Alternating lead-reviewer roles*, in which one participant leads each step while the other reviews, with roles alternating across steps; (3) *Structured inter-step reflection prompts*, which pause the flow at each transition and

present a joint question both participants must answer; (4) *Contribution awareness display*, a real-time visual indicator of each participant’s contributions; and (5) *Dual observation logging*, in which both participants independently record observations during testing and the system surfaces agreements and divergences. These supports are specified and under active development.

Together, these elements create a collaboration architecture that occupies a middle ground between fully symmetric access and fixed role assignment [10, 13], structuring joint specification while preserving shared ownership of the design artifact.

4 Toward Family-Centered CollabPEAS

CollabPEAS was designed for peer dyads in formal educational settings. Extending it to family contexts does not mean starting over, but it does raise design questions worth thinking through.

4.1 Knowledge Asymmetry as a Collaborative Resource

In a peer dyad, both participants start at roughly the same level of AI literacy. In a parent-child pair, that is rarely the case, and the asymmetry can run in opposite directions depending on what the task asks for. A parent is often better placed to define performance measures, because they have more developed intuitions about household goals, trade-offs, and what “good enough” means in a domestic context. A child may be more confident with sensor selection and block-coding, where the interaction patterns resemble gaming or creative computing tools they already use. CollabPEAS’s alternating lead-reviewer structure fits this well. Assigning the parent as lead for the earlier steps around performance measure and environment, then handing the lead to the child during block-coding and debugging, could turn the knowledge gap into a mutual teaching structure rather than a source of friction. The family dyad is not a peer pair with an imbalance to fix; it is a pair whose different strengths could, with the right structure, produce richer specification work than either person would do alone.

4.2 The Home as a Specification Context

CollabPEAS currently uses a generic apartment as the agent’s environment. In a family-centered version, this becomes a resource: the home the family actually inhabits provides a deeply meaningful specification context. Asking a parent and child to specify a cleaning robot for their own kitchen, with their own criteria for what “clean enough” means and their own layout constraints, grounds the PEAS framework in personally relevant design choices rather than generic ones. This aligns with research showing that AI literacy activities are more engaging and conceptually productive when they connect to learners’ lived contexts [4]. It also raises a design question that CollabPEAS does not yet address: how to support families in transferring the specification reasoning they develop in the VR environment back to their understanding of real AI systems they already use at home.

4.3 Negotiating AI Governance as a Family Practice

The agree-before-placing mechanism and inter-step reflection prompts in CollabPEAS were designed to prevent one learner from dominating the specification process. In a family context, these features carry additional weight. The prompts (“Do you both agree these sensors give the robot enough information to do its job?”) mirror the kinds of decisions families face when adopting AI tools at home, including decisions about whose preferences the agent should optimize, what data it should collect, and when it should defer to a person. Practiced inside a learning experience, these moments of explicit negotiation could build habits for the kind of ongoing family AI governance that the workshop

identifies as one of its central concerns [9]. The collaboration supports in CollabPEAS are not only pedagogical tools; in a family context, they also function as rehearsal for shared decision-making about AI at home.

5 Discussion and Open Questions

Thinking about CollabPEAS in a family context opens up several design questions worth bringing to the workshop.

How should role structure adapt to family dynamics? The alternating lead-reviewer structure assumes roughly equal authority between participants. In parent-child pairs, authority asymmetry is real and varies across task type and child age. Should the system leave role negotiation to the family, or should it offer family-sensitive role structures that evolve as the child demonstrates competence?

What is the right level of scaffolding for intergenerational pairs? The current system’s inter-step reflection prompts assume both participants can engage with the same level of conceptual abstraction. For a six-year-old and a parent, the prompts may need to be scaffolded differently than for a teenager and a parent. Adaptive scaffolding that responds to demonstrated understanding rather than fixed step-level checkpoints may be needed.

How do families transfer in-system learning to out-of-system AI encounters? The most important question for family-centered AI literacy may not be what families learn about a simulated cleaning robot, but whether the specification reasoning they practice carries over to conversations about the AI systems they actually encounter at home: the voice assistant in the kitchen, the recommendation engine on the tablet, the navigation system in the car. Designing for this transfer requires connecting the VR learning environment to the family’s real-world AI encounters in ways that CollabPEAS does not yet address.

6 Conclusion

CollabPEAS shows that PEAS-based agent specification can be made accessible and engaging for non-technical learners through collaborative VR. The features that structure negotiation and shared ownership between partners become, in a family setting, features that also structure how knowledge is transferred across generations and how AI governance is practiced at home. These are directions worth pursuing, and the questions raised here are offered as a starting point for that work.

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